STAT 301: Mathematical Statistics and Modeling I

Instructor: Dr. Erin Blankenship (she/her)

email: `erin.blankenship@unl.edu`

Lecture: 12:00-12:50 Monday/Wednesday/Friday, 272 Plant Sciences Hall

Office Hours: 9:00-10:00 Monday/Wednesday/Friday or by appointment; 343B Hardin Hall, North Wing

Required Materials: No book is required.

Reference: *Mathematical Statistics with Applications*, 7th edition (Wackerly, Mendenhall and Scheaffer; 2008)

Prerequisites: MATH 314, STAT 212, STAT 262

Course Description: Essential statistical theory and methods for professional statistical practice. Broad statistical topics include estimation and hypothesis testing, elementary Bayesian concepts, multiple linear regression, linear mixed effects models, analysis of variance (ANOVA), logistic regression, Poisson regression, and nonparametric methods.

Course Goals:

- Understand different methods of obtaining point estimates, including maximum likelihood, method of moments, and Bayesian estimation
- Understand that some estimators are better than others, and be able to evaluate estimators
- Understand the theory underlying hypothesis tests, including errors, power, size, and level of tests
- Be able to derive hypothesis tests for simple situations from first principles, including likelihood ratio tests and most powerful tests
- Be able to derive interval estimates by inverting tests and using pivotal quantities

Grading:

Assignment(s)	Contribution to Final Grade
Homework	25%
Exam 1	25%
Exam 2	25%
Final Project	25%

Grading Scale:

Grade	Final Percentage Range
A	94.0-100
A-	90.0-93.99
B+	88.0-89.99
B	84.0-87.99
B-	80.0-83.99
C+	78.0-79.99
C	74.0-77.99
C-	70.0-73.99
D+	68.0-69.99
D	64.0-67.99
D-	60.0-63.99
F	<60.0

Course Expectations: In this course, you are expected to have professional behavior. You are expected to attend all class meetings, be curious, ask questions, seek opportunities to learn, and be open and responsive to constructive feedback. In addition:

- Be an active participant–statistics is not a spectator sport!
- Be committed, take your work seriously
- Engage with the in-class activities and labs
- Help others–if you understand the material being discussed, practice your mentoring skills. This does not means sharing answers, but instead helping others understand the concepts.
- Complete any assigned readings.

You are also expected to exhibit a professional demeanor (language, attitude) toward others. Disagreement during discussions is welcome and often productive in developing a deeper understanding of the concepts being discussed. However, disagreement does not warrant yelling or disrespectful language or behavior. Unprofessional behavior will not be tolerated, and appropriate actions will be taken to prevent future occurrences.

Recording of class-related activity: I invite all of you to join me in actively creating and contributing to a positive, productive, and respectful classroom culture. Each student contributes to an environment that shapes the learning process. Any work and/or communication that you are privy to as a member of this course should be treated as the intellectual property of the speaker/creator, and is not to be shared outside the context of this course.

Students may not make or distribute screen captures, audio/video recordings of, or livestream, any class-related activity, including lectures and presentations, without express prior written consent from me or an approved accommodation from Services for Students with Disabilities. If you have (or think you may have) a disability such that you need to record or tape class-related activities, you should contact Services for Students with Disabilities. If you have an accommodation to record class-related activities, those recordings may not be shared with any other student, whether in this course or not, or with any other person or on any other platform.

Homework: Approximately 8-10 homework assignments will be made over the course of the semester. The only way to learn statistics is to practice working problems, and homework is therefore an essential part of the course. Bear in mind that homework is for your benefit, not the instructor's.

Homework may be submitted either on paper or through Canvas. If you opt to submit via Canvas, it must be submitted as a single pdf (not multiple files), and any scans must be high quality. If I cannot read it, it will not be graded. Homework will be graded partially on completeness and partially on accuracy.

Exams: Two exams will be given during the course of the semester, tentatively scheduled on the dates in the course outline. The in-class exams are closed book and notes. The exams will evaluate your understanding of the material, as well as your ability to synthesize and transfer that knowledge to other scenarios and situations; questions will assess conceptual understanding as opposed to mere memorization.

Calculators: We will be using a great deal of calculus in this course. While I realize that many calculators will do the calculus we'll use, such calculators will **not** be permitted on in-class exams. Don't get too reliant on them for homework!

Final Project: A final group project will take the place of a final exam. The project will include a presentation during the last week of class and a final written report. More details on the project will be distributed later.

Instructional Continuity: If in-person classes are canceled, you will be notified of the instructional continuity plan for this class through Canvas.

AI and Explainability Policy: (adapted from Dr. Vanderplas)

- Any use of generative AI must be disclosed in an appendix to your submission - this includes brainstorming, editing, using AI as spell-check/grammar-check, and so on. You must document the following:
  - the version of the generative AI used
  - the full sequence of prompts and responses
  - any additional inputs you provided to the AI system
  - a "diff" between the AI responses and your submission, showing exactly what was generated by the AI system and what you changed.

It may be useful to leverage AI tools to ensure that your work conforms to grammar and style guidelines, but I very highly discourage the use of generative AI for content or code (I've seen it generate incorrect code too many times).

- You must be able to explain any work you turn in, including code. If you cannot explain the logic behind your approach as well as how it works in practice, then you will not receive credit for your submission.

Department Grade Appeal Policy: The Department of Statistics grade appeal policy

Academic Integrity: You are encouraged to work together on problems and exercises, but the work you turn in must be your own (unless the assignment specifically states otherwise). Work on exams must be your own. University policy will be followed in cases of academic dishonesty: In cases where an instructor finds that a student has committed any act of academic dishonesty, the instructor may in the exercise of his or her professional judgment impose an academic sanction as severe as giving the student a failing grade in the course. Before imposing an academic sanction the instructor shall first attempt to discuss the matter with the student. If deemed necessary by either the instructor or the student, the matter may be brought to the attention of the student's major advisor, the instructor's department chairperson or head, or the dean of the college in which the student is enrolled. For additional details see the department policy.

Services for Students with Disabilities: The University strives to make all learning experiences as accessible as possible. If you anticipate or experience barriers based on your disability (including mental health, chronic or temporary medical conditions), please let your instructor know immediately so that you can discuss options privately. To establish reasonable accommodations, your instructor may request that you register with Services for Students with Disabilities. If you are eligible for services and register with the office, make arrangements with your instructor as soon as possible to discuss your accommodations so they can be implemented in a timely manner. SSD is located in 117 Louise Pound Hall and can be reached at 402-472-3787.

University Policies: All university-wide course policies.

TENTATIVE **Course Outline**

Week	Dates	Notes	Topics
1	24-28 August		Order statistics
2	31 Aug -4 Sept	No class on 4 Sept	Order statistics
3	7-11 September		Point estimation; method of moments
4	11-18 September		Maximum likelihood estimation
5	21-25 September		Evaluating estimators
6	28 Sept - 2 Oct		Sufficiency, completeness, and best estimation
7	5-9 October		Best estimators; Review and **Exam 1**
8	12-16 October		Bayes estimators; intro to hyp testing
9	19-23 October	No class on 19 Oct	Intro to hyp testing
10	26-30 October		Likelihood ratio tests
11	2-6 November		NP Lemma; MLR; Karlin-Rubin
12	9-13 November		p-values; intro to interval estimates
13	16-20 November		Inverting tests; pivotal quantities
14	23-27 November	No class on 25 or 27 Nov	Bayesian intervals
15	30 Nov - 4 Dec		Catch-up; Review and **Exam 2**
16	7-11 December		Delta-method; **Final project presentations on 11 December**
Finals Week		No final exam	**Final project papers due Monday, 14 December at 9:00 am**